

MTH 732 Topology - Homework 3

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Assignment 1. Show that $\pi_1(\mathbb{RP}^2) = \mathbb{Z}_2$. Describe the generator of this group.

Proof. Recall that $\mathbb{RP}^2$ can be constructed as S^2 with antipodal points identified. Let $p : S^2 \rightarrow \mathbb{RP}^2$ be the projection which sends $x \mapsto \{x, -x\}$. Then for any point x , a neighborhood U of $\{x, -x\}$ in $\mathbb{RP}^2$ lifts to exactly two open sets in S^2 , namely a neighborhood of x and a neighborhood of $-x$. Furthermore, the set U can be chosen to be small enough so that its lifts are disjoint, and each one is naturally homeomorphic to U . Since S^2 is simply connected, this makes it a universal cover of $\mathbb{RP}^2$. As such, since each fiber over p consists of exactly two points, we can conclude that $\pi_1(\mathbb{RP}^2) \cong \mathbb{Z}_2$ since $\mathbb{Z}_2$ is the only two-element group up to isomorphism.

Now let $\alpha : [0, 1] \rightarrow S^2$ be a path from x to $-x$ for some $x \in S^2$. Then $p \circ \alpha \in \pi_1(\mathbb{RP}^2)$, and in particular $p \circ \alpha$ is nontrivial since if it were null-homotopic, the corresponding path homotopy would lift to a path homotopy in S^2 that ends in a constant path, which is impossible since endpoints need to remain constant in a path-homotopy, and the endpoints x and $-x$ are distinct. $\square$

Assignment 2. Show that any map $f : \mathbb{RP}^2 \rightarrow \mathbb{T}^2$ is homotopically trivial.

Proof. Given $f : \mathbb{RP}^2 \rightarrow \mathbb{T}^2$, we have the induced group homomorphism $f_* : \mathbb{Z}_2 \rightarrow \mathbb{Z}^2$. Any such map must be trivial since it must send any generator of $\mathbb{Z}_2$, which has order 2, to an element in $\mathbb{Z}^2$ whose order divides two. Since there are no elements of order 2 in $\mathbb{Z}^2$, this must be an element of order 1, namely the identity.

Let $p : \mathbb{R}^2 \rightarrow \mathbb{T}^2$ be the universal covering map. Then since the trivial group is a subgroup of every group, in particular $\text{im } f_* \subset \text{im } p_*$, so we have a lift $\tilde{f} : \mathbb{RP}^2 \rightarrow \mathbb{R}^2$. Since $\mathbb{R}^2$ is contractible, we have a homotopy $H : \mathbb{RP}^2 \times [0, 1] \rightarrow \mathbb{R}^2$ from $\tilde{f}$ to a constant map. Then the composition $p \circ H$ is itself a homotopy from f to the constant map, as desired. $\square$

Assignment 3. If $f : S^1 \rightarrow S^1$ satisfies $|f(z) - z| < 1$ for all $z \in S^1$, show that f is surjective.

Proof. If f is not surjective, then there is a point $x \in S^1$ such that $x \notin \text{im } f$. If this is the only missing point, then $\text{im } f$ is homeomorphic to an open interval. We know that $\text{im } f$ has one connected component since S^1 has one component, so we must indeed have $\text{im } f$ being homeomorphic to an open interval, say $(0, 1)$. Since this interval is contractible, f is null-homotopic.

This would seem to imply that the degree of f is 0. However, the condition $|f(z) - z| < 1$ implies that the degree of f is 1, since if it were $n \in \mathbb{Z} \setminus \{1\}$, we would have some point being sent to its antipodal point, which would be a distance of greater than 1. This is a contradiction. $\square$

Assignment 4. Show that the system of equations

$$\begin{cases} \cos(x^2y) - 4x(1+y^2) = 0 \\ \sin(e^{x+2y})|x-y| - 5y = 0 \end{cases}$$

has a solution.

Proof. We can rewrite this system of equations as

$$\begin{cases} \frac{\cos(x^2y)}{4(1+y^2)} = x \\ \frac{\sin(e^{x+2y})|x-y|}{5} = y \end{cases}.$$

Now this reduces to finding a fixed point of $f : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ defined by

$$f(x, y) = \left(\frac{\cos(x^2y)}{4(1+y^2)}, \frac{\sin(e^{x+2y})|x-y|}{5} \right).$$

Since $\text{im } f \subset D^2$, $f|_{D^2} : D^2 \rightarrow D^2$ (where D^2 has radius 1) has the same fixed points (if any), we conclude by the Brouwer fixed point theorem that there is one. $\square$

Assignment 5. Consider the group G with presentation

$$G = \langle x, y : xyx = yxy \rangle.$$

Show that G is not Abelian by finding a suitable homomorphism from G into S_3 , the symmetric group on three symbols.

Proof. Define $f : G \rightarrow S_3$ by $f(x) = (1\ 2)$ and $f(y) = (1\ 3)$. Then indeed,

$$\begin{aligned} f(xyx) &= f(x)f(y)f(x) \\ &= (1\ 2)(1\ 3)(1\ 2) \\ &= (2\ 3) \\ &= (1\ 3)(1\ 2)(1\ 3) \\ &= f(y)f(x)f(y) \\ &= f(yxy), \end{aligned}$$

so this satisfies the relation and is thus a homomorphism. For G to be Abelian, we would in particular have $xy = yx$ and thus $f(x)f(y) = f(y)f(x)$. Since $(1\ 2)(1\ 3) = (1\ 3\ 2)$ and $(1\ 3)(1\ 2) = (1\ 2\ 3)$, this is not the case, so G is not abelian. $\square$